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Combustor

Abstract

A combustor includes: a housing having an annular shape of which one end side is open and of which the other end side is closed; at least one introduction portion that introduces a fuel and an oxidizing gas into the housing to generate a tubular flow; and an ignition unit that ignites the fuel introduced into the housing. The ignition unit includes a discharge electrode and a ground electrode. A space that the fuel and the oxidizing gas reach is provided between the discharge electrode and the ground electrode.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/JP2021/000841, filed Jan. 13, 2021, claiming priorities to Japanese Patent Application No. 2020-018145, filed Feb. 5, 2020 and Japanese Patent Application No. 2020-165666, filed Sep. 30, 2020.

TECHNICAL FIELD

(2) The present disclosure relates to a combustor.

BACKGROUND ART

(3) As a combustor of the related art, for example, a technique disclosed in Patent Literature 1 has been known. The combustor disclosed in Patent Literature 1 includes a combustion chamber having a tubular shape in which a tip is open and serves as a discharge port for combustion exhaust gas, a plurality of nozzles attached to the vicinity of a rear end of the combustion chamber to inject a fuel gas and an oxygen-containing gas into the combustion chamber, and an ignition spark plug that is attached to the rear end of the combustion chamber and that is caused to fire a spark in the combustion chamber by an igniter and a power source. The nozzles are provided to inject the fuel gas and the oxygen-containing gas in a tangential direction of an inner peripheral surface of the combustion chamber. The ignition spark plug is disposed between a tube axis of the combustion chamber and a position of r/2 (r: radius of the combustion chamber).

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Unexamined Patent Publication No. 2004-93114

SUMMARY OF INVENTION

Technical Problem

(5) However, in the technique of the related art, a mixed gas of the fuel gas and the oxygen-containing gas is ignited only in the vicinity of a radially center portion of the combustion chamber having a tubular shape. Therefore, it is necessary to locally adjust the flow speed and the air-fuel ratio of the fuel gas and the oxygen-containing gas according to the ignition position, and it is difficult to secure the ignition stability of the fuel gas. The ignition stability referred to here means reliably performing ignition within a desired time.

(6) An object of the present disclosure is to provide a combustor capable of improving the ignition stability of a fuel.

Solution to Problem

(7) A combustor according to one aspect of the present disclosure includes: a housing having a circular tubular shape in which one end forms an open end that is open, and in which a closed wall is provided at the other end; at least one introduction portion that introduces a fuel and an oxidizing gas into the housing to generate a tubular flow; and an ignition unit that ignites the fuel introduced into the housing. The housing is grounded to function as a ground electrode. The ignition unit includes a discharge electrode terminal disposed in a region including an inside on a closed wall side of the housing, to function as a discharge electrode, and a voltage supply unit that supplies a voltage to the discharge electrode terminal, and generates a discharge between a tip portion of the

discharge electrode terminal and the housing to ignite the fuel.
(8) In such a combustor, when the introduction portion introduces the fuel and the oxidizing gas into the housing having a circular tubular shape to generate a tubular flow, the fuel and the oxidizing gas form a tubular flow and flow inside the housing in a swirling manner. At this time, the fuel and the oxidizing gas flowing toward the closed wall of the housing hit the closed wall and flow while changing the direction. In that state, when the voltage supply unit supplies a voltage to the discharge electrode terminal, a discharge is generated between the tip portion of the discharge electrode terminal and the grounded housing, so that the fuel is ignited and combusted to generate combustion gas including a tubular flame. Then, the combustion gas flows inside the housing toward the open end of the housing and is discharged from the open end. Here, the discharge is generated in a wide range between the tip portion of the discharge electrode terminal and the housing. In addition, since the closed wall side of the housing is used as the ground electrode, the electrode area is substantially increased. In addition, a dielectric breakdown is directed from the discharge electrode terminal toward the housing, and discharge paths are radially formed at 360 degrees. Therefore, the structure is unlikely to be affected by a local fluctuation in gas flow and a local surface condition of the housing, and the ignition stability of the fuel is improved. In addition, the discharge electrode terminal is disposed in the region including the inside on the closed wall side of the housing. Therefore, a discharge is generated at a position away from the open end of the housing, the open end forming the discharge port for the combustion gas. Accordingly, combustion is stabilized inside the housing.
(9) The discharge electrode terminal may be attached to the closed wall via an insulating body. In such a configuration, the discharge electrode terminal is unlikely to interfere with a tubular flow of the fuel and the oxidizing gas. Therefore, the fuel and the oxidizing gas easily flow inside the housing in a swirling manner, so that the ignition stability of the fuel is more improved.
(10) The discharge electrode terminal may be disposed at a radially center portion of the housing inside the housing. In such a configuration, since a distance from the tip portion of the discharge electrode terminal to the housing is equal over the whole circumference of the housing, a discharge is uniformly generated between the tip portion of the discharge electrode terminal and the housing along a circumferential direction of the housing. Therefore, the ignition stability of the fuel is more improved.
(11) The tip portion of the discharge electrode terminal may be located between an end on the closed wall side of the introduction portion and the closed wall. Even in such a configuration, the discharge electrode terminal is unlikely to interfere with a tubular flow of the fuel and the oxidizing gas. Therefore, the fuel and the oxidizing gas easily flow inside the housing in a swirling manner, so that the ignition stability of the fuel is more improved. In addition, since a discharge is generated at a position sufficiently away from the open end of the housing, combustion is more stabilized inside the housing.
(12) According to one aspect of the present disclosure, there is provided a combustor that combusts a fuel mixed with an oxidizing gas, to generate combustion gas, the combustor including: a housing having a circular tubular shape of which one end side is open and of which the other end side is closed and through which the fuel, the oxidizing gas, and the combustion gas flow in an axial direction; at least one introduction portion that introduces the fuel and the oxidizing gas into the housing to generate a tubular flow; and an ignition unit that includes an ignition plug disposed on the other end side of the housing and that ignites the fuel introduced into the housing, to generate the combustion gas including a tubular flame. The ignition plug includes an insulator, a center electrode that is provided at a tip of a center pole supported by the insulator, to function as a discharge electrode, a metal shell having a cylindrical shape and being disposed around the insulator, and a ground electrode integrated with the metal shell. The ignition unit ignites the fuel by supplying a voltage to the center electrode and thus generating a discharge in a space between the center electrode and the ground electrode. The center electrode protrudes from a tip surface of

the insulator. The ground electrode is disposed outside the center electrode in a radial direction of the metal shell so as not to overlap the center electrode in an axial direction of the metal shell.

(13) In such a combustor, when the introduction portion introduces the fuel and the oxidizing gas into the housing having a circular tubular shape, the fuel and the oxidizing gas form a tubular flow and flow inside the housing toward the ignition plug. In that state, when a voltage is supplied to the center electrode, a discharge is generated in the space between the center electrode and the ground electrode, so that the fuel is ignited and combusted to generate combustion gas including a tubular flame. The combustion gas is discharged from one end of the housing.

(14) Here, the ground electrode is integrated with the metal shell having a cylindrical shape and being disposed around the insulator, and is disposed outside the center electrode in the radial direction of the metal shell so as not to overlap the center electrode in the axial direction of the metal shell. For this reason, the ground electrode is prevented from interfering with a flow of the fuel and the oxidizing gas to the space between the center electrode and the ground electrode even when a flow of the fuel and the oxidizing gas toward the ignition plug is generated in a radially inner (center side) region of the housing inside the housing by a tubular flow of the fuel and the oxidizing gas. Therefore, the fuel and the oxidizing gas easily flow into the space between the center electrode and the ground electrode, so that a discharge generated in the space between the center electrode and the ground electrode easily acts on a mixture gas of the fuel and the oxidizing gas. In addition, after the fuel is ignited by the discharge, the flame burns and spreads to the surrounding mixed gas, but since the mixed gas flows as a tubular flow to the one end side of the housing in the axial direction of the metal shell, the tubular flame expands in the axial direction. At this time, the ground electrode is disposed not to overlap the center electrode in the axial direction of the metal shell. For this reason, when the fuel is ignited, heat is unlikely to be taken away from the tubular flame by the ground electrode and the tubular flame easily expands. As described above, the ignition and combustion operation of the fuel is stabilized.

(15) The ground electrode may have a cylindrical shape and be integrated with the metal shell so as to be disposed around the insulator. In such a configuration, a structure of the existing ground electrode can be used.

(16) A tip of the center electrode may be located on the one end side of the housing with respect to the ground electrode. In such a configuration, the fuel and the oxidizing gas easily flow from a center electrode side to a ground electrode side. Therefore, the fuel and the oxidizing gas more easily flow into the space between the center electrode and the ground electrode.

(17) The center electrode may include a protrusion portion protruding in a radial direction of the ground electrode with respect to a peripheral edge of the tip surface of the insulator. A side end of the protrusion portion may be located in a region outside the tip surface of the insulator in the radial direction of the ground electrode and inside the ground electrode in the radial direction of the ground electrode. In such a configuration, when a voltage is supplied to the center electrode, a discharge is generated in a space between the protrusion portion and the ground electrode. Therefore, the discharge is generated in the space separated from the insulator, so that heat caused by the ignition of the fuel is unlikely to be taken away by the insulator. Accordingly, the ignition and combustion operation of the fuel is more stabilized.

(18) The center electrode may have a circular shape. A diameter of the center electrode may be larger than a diameter of the tip surface of the insulator and smaller than a diameter of the ground electrode. The protrusion portion may be provided at a peripheral edge portion of the center electrode and have an annular shape. In such a configuration, a discharge can be generated in the space between the protrusion portion and the ground electrode over the whole circumference of the center electrode. In addition, since the shape of the center electrode is a circular shape, the center electrode can be easily manufactured.

(19) A tip portion of the ground electrode may be provided with a projection protruding to the one end side of the housing. In such a configuration, when a voltage is supplied to the center electrode,

a discharge is generated in a space between the center electrode and the projection. Therefore, the discharge is generated in the space separated from the insulator, so that heat caused by the ignition of the fuel is unlikely to be taken away by the insulator. Accordingly, the ignition and combustion operation of the fuel is more stabilized.
(20) A tip of the projection may be located on the one end side of the housing with respect to the center electrode. In such a configuration, when a voltage is supplied to the center electrode, a discharge is generated in a space between the center electrode and a side surface of the projection. Therefore, the discharge is generated in the space sufficiently separated from the insulator, so that heat caused by the ignition of the fuel is more unlikely to be taken away by the insulator. Accordingly, the ignition and combustion operation of the fuel is even more stabilized.
(21) The ground electrode may include an annular portion attached to the metal shell, and an erected portion provided on a tip surface of the annular portion to extend to the one end side of the housing. In such a configuration, when a voltage is supplied to the center electrode, a discharge is generated in a space between the center electrode and the erected portion. Therefore, the discharge is generated in the space separated from the insulator, so that heat caused by the ignition of the fuel is unlikely to be taken away by the insulator. Accordingly, the ignition and combustion operation of the fuel is more stabilized.
(22) A plurality of the erected portions may be provided on the tip surface of the annular portion. The ground electrode may further include a connection portion connecting tips of the plurality of erected portions. A tip of the connection portion may be located on the one end side of the housing with respect to the center electrode. A distance between the center electrode and the erected portion may be shorter than a distance between the center electrode and the connection portion. In such a configuration, the tips of the plurality of erected portions are connected by the connection portion, thereby leading to an increase in the strength of the erected portions. In addition, when a voltage is supplied to the center electrode, a discharge is generated in a space between the center electrode and the erected portions, and a discharge is not generated in a space between the center electrode and the connection portion.
(23) The metal shell may be fixed to the housing. The ground electrode may include a protrusion provided on an inner peripheral surface of the housing to protrude inward in a radial direction of the housing toward the center electrode. In such a configuration, when a voltage is supplied to the center electrode, a discharge is generated in a space between the center electrode and the protrusion. At this time, when the protrusion is disposed away from the insulator, heat caused by the ignition of the fuel is unlikely to be taken away by the insulator. Accordingly, the ignition and combustion operation of the fuel is more stabilized.
Advantageous Effects of Invention
(24) According to the present disclosure, it is possible to improve the ignition stability of the fuel.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a configuration view illustrating a tubular flame burner according to a first embodiment of the present disclosure.
(2) FIG. **2** is a cross-sectional view taken along line II-II of FIG. **1**.
(3) FIG. **3** is a view illustrating a state where a plasma caused by a discharge is formed inside a housing illustrated in FIG. **1**.
(4) FIG. **4** is a view illustrating a state where a plasma caused by a discharge is formed inside the housing in a modification example of the tubular flame burner illustrated in FIG. **3**.
(5) FIG. **5** is a cross-sectional view illustrating a combustor according to a second embodiment of the present disclosure.

(6) FIG. **6** is a cross-sectional view taken along line II-II of FIG. **5**.
(7) FIG. **7** is a side cut-away view of an ignition plug illustrated in FIG. **5**.
(8) FIG. **8** is a schematic front view of the ignition plug illustrated in FIG. **7**.
(9) FIG. **9** is a side cut-away view illustrating a mode in which a discharge is generated in a space between a center electrode and a ground electrode in the ignition plug illustrated in FIG. **7**.
(10) FIG. **10** is a cross-sectional view illustrating a combustor as a comparative example.
(11) FIG. **11** is a side cut-away view illustrating an ignition plug in a combustor according to a third embodiment of the present disclosure.
(12) FIG. **12** is a schematic front view of the ignition plug illustrated in FIG. **11**.
(13) FIG. **13** is a schematic front view of an ignition plug including a modification example of a center electrode illustrated in FIG. **12**.
(14) FIG. **14** is a side cut-away view illustrating an ignition plug in a combustor according to a fourth embodiment of the present disclosure.
(15) FIG. **15** is a schematic front view of the ignition plug illustrated in FIG. **14**.
(16) FIG. **16** is a side cut-away view of an ignition plug including a modification example of a ground electrode illustrated in FIG. **14**.
(17) FIG. **17** is a side cut-away view illustrating an ignition plug in a combustor according to a fifth embodiment of the present disclosure.
(18) FIG. **18** is a side cut-away view illustrating an ignition plug in a combustor according to a sixth embodiment of the present disclosure.
(19) FIG. **19** is a side cut-away view of an ignition plug including a modification example of a ground electrode illustrated in FIG. **18**.
(20) FIG. **20** is a side cut-away view illustrating an ignition plug in a combustor according to a seventh embodiment of the present disclosure.
(21) FIG. **21** is a side cut-away view illustrating an ignition plug in a combustor according to an eighth embodiment of the present disclosure.

DESCRIPTION OF EMBODIMENTS

(22) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. Incidentally, in the drawings, the same or equivalent elements are denoted by the same reference signs, and duplicated descriptions will be omitted.

(23) [First Embodiment] FIG. **1** is a configuration view illustrating a tubular flame burner according to one embodiment of the present disclosure. In FIG. **1**, a tubular flame burner (combustor) **100** of the present embodiment is a device that mixes ammonia gas (NH.sub.3 gas) that is a fuel and air that is an oxidizing gas to combust the ammonia gas.

(24) The tubular flame burner **100** includes a housing **20** having a circular tubular shape, two ammonia gas introduction portions **30** that introduce the ammonia gas into the housing **20**, two air introduction portions **400** that introduce the air into the housing **20**, and an ignition unit **500** that ignites the ammonia gas introduced into the housing **20**.

(25) The housing **20** includes a circular cylindrical portion **20***b* of which both ends are open. One end of the housing **20** forms an open end **20***a* that is open. The open end **20***a* opens one end of the circular cylindrical portion **20***b* to the atmosphere and serves as a discharge port for combustion gas to be described later. A closed wall **60** having a circular plate shape is provided at the other end of the housing **20**. The closed wall **60** closes the other end of the circular cylindrical portion **20***b*. The closed wall **60** is firmly fixed to the other end portion of the circular cylindrical portion **20***b*. The housing **20** is made of a metal material having conductivity (for example, stainless steel).

(26) The ammonia gas introduction portion **30** and the air introduction portion **400** are introduction portions provided at a center portion in an axial direction of the housing **20** on an outer peripheral surface of the housing **20**, or on a closed wall **60** side with respect to the center portion. As illustrated in FIG. **2**, the ammonia gas introduction portions **30** and the air introduction portions **400** are alternately disposed at equal intervals along a circumferential direction of the housing **20**.

The ammonia gas introduction portion **30** introduces the ammonia gas into the housing **20** in a tangential direction of an inner peripheral surface **20***c* of the housing **20**. Namely, the ammonia gas introduction portion **30** introduces the ammonia gas into the housing **20** to generate a tubular flow. The air introduction portion **400** introduces the air into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**. Namely, the air introduction portion **400** introduces the air into the housing **20** to generate a tubular flow. The ammonia gas introduction portions **30** and the air introduction portions **400** may be integrally formed with the housing **20** or may be configured separately from the housing **20** and fixed to the housing **20**.
(27) The ignition unit **500** includes a discharge electrode terminal **70** disposed inside the housing **20**, an ignitor **80** that applies a high voltage to the discharge electrode terminal **70** to ignite the ammonia gas, and a power source **90** that turns on and off the ignitor **80**. The ignitor **80** and the power source **90** form a voltage supply unit **140** that supplies a voltage to the discharge electrode terminal **70**.
(28) The discharge electrode terminal **70** is disposed in a region including an inside on the closed wall **60** side of the housing **20**. The inside on the closed wall **60** side of the housing **20** indicates a region on the closed wall **60** side inside the housing **20** with respect to a center in the axial direction (longitudinal direction) of the housing **20**.
(29) Specifically, the discharge electrode terminal **70** is attached to the closed wall **60** via an insulating body **110** to penetrate through the closed wall **60** of the housing **20**. The insulating body **110** is made of an insulating material (for example, ceramic) having pressure resistance and heat resistance.
(30) The discharge electrode terminal **70** is disposed at a radially center portion of the housing **20** inside the housing **20** to extend in the axial direction of the housing **20**. A part of the discharge electrode terminal **70** is disposed inside the housing **20**, and the remaining portion of the discharge electrode terminal **70** is disposed outside the housing **20**.
(31) A tip portion **70***a* of the discharge electrode terminal **70** is located between each of a front end **30***a* of the ammonia gas introduction portion **30** and a front end **400***a* of the air introduction portion **400** and the closed wall **60** of the housing **20**. Here, the tip portion **70***a* of the discharge electrode terminal **70** is located between each of a rear end **30***b* of the ammonia gas introduction portion **30** and a rear end **400***b* of the air introduction portion **400** and the closed wall **60**.
(32) The tip portion **70***a* of the discharge electrode terminal **70** is an end portion of both end portions of the discharge electrode terminal **70**, which is disposed inside the housing **20** (end portion on an open end **20***a* side). The front end **30***a* of the ammonia gas introduction portion **30** and the front end **400***a* of the air introduction portion **400** correspond to ends of the ammonia gas introduction portion **30** and the air introduction portion **400** on the open end **20***a* side of the housing **20**. The rear end **30***b* of the ammonia gas introduction portion **30** and the rear end **400***b* of the air introduction portion **400** correspond to ends of the ammonia gas introduction portion **30** and the air introduction portion **400** on the closed wall **60** side of the housing **20**.
(33) The discharge electrode terminal **70** is connected to the ignitor **80** via an electric wire **120**. A pulse voltage from the ignitor **80** is supplied to the discharge electrode terminal **70** via the electric wire **120**. The discharge electrode terminal **70** functions as a discharge electrode. The housing **20** is connected to a ground line (GND line) of the ignitor **80** via an electric wire **130**. Therefore, the housing **20** is grounded. The housing **20** functions as a ground electrode. A space that the ammonia gas and the air reach is provided between the discharge electrode terminal **70** and the housing **20**.
(34) The ignition unit **500** ignites the ammonia gas by applying a high voltage to the discharge electrode terminal **70** and thus generating a discharge between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20**. At this time, as illustrated in FIG. **3**, a plasma P is formed between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20** by the discharge generated between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20**.

(35) In the tubular flame burner **100** as described above, when the ammonia gas introduction portions **30** introduce the ammonia gas into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**, and the air introduction portions **400** introduce the air into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**, the ammonia gas and the air form a tubular flow and are mixed and flow inside the housing **20** in a swirling manner. At this time, a mixed gas of the ammonia gas and the air flows inside the housing **20** toward the open end **20***a* of the housing **20**, and flows inside the housing **20** toward the closed wall **60** of the housing **20** to hit the closed wall **60** and then to flow while changing the direction.
(36) In that state, when the power source **90** is turned on, the ignitor **80** applies a high voltage to the discharge electrode terminal **70**, a discharge is generated between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20**, and the plasma P is formed between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20** by the discharge. Then, the ammonia gas is ignited and combusted to form a tubular flame, so that high-temperature combustion gas is generated.
(37) Here, since the ammonia gas and the air are introduced into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**, the flow speed of the ammonia gas and the gas is higher on a radially outer side of the housing **20** than on a radially inner side (center side) of the housing **20**. However, since a discharge is generated in a wide range between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20**, regardless of the flow speed and the air-fuel ratio of the ammonia gas and the air, the ammonia gas is easily ignited and combusted inside the housing **20**.
(38) The high-temperature combustion gas obtained by the combustion of the ammonia gas flows inside the housing **20** toward the open end **20***a* of the housing **20**. Then, the combustion gas is discharged from the open end **20***a* forming the discharge port.
(39) As described above, in the present embodiment, when the ammonia gas introduction portions **30** introduce the ammonia gas and the air into the housing **20** having a circular tubular shape to generate a tubular flow, the ammonia gas and the air form a tubular flow and flow inside the housing **20** in a swirling manner. At this time, the ammonia gas and the air flowing toward the closed wall **60** of the housing **20** hit the closed wall **60** and flow while changing the direction. In that state, when the voltage supply unit **140** supplies a voltage to the discharge electrode terminal **70**, a discharge is generated between the tip portion **70***a* of the discharge electrode terminal **70** and the grounded housing **20**, so that the ammonia gas is ignited and combusted to generate the combustion gas including a tubular flame. Then, the combustion gas flows inside the housing **20** toward the open end **20***a* of the housing **20** and is discharged from the open end **20***a*. Here, the discharge is generated in a wide range between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20**. In addition, since the closed wall **60** side of the housing **20** is used as the ground electrode, the electrode area is substantially increased. In addition, a dielectric breakdown is directed from the discharge electrode terminal **70** toward the housing **20**, and discharge paths are radially formed at 360 degrees. Therefore, the structure is unlikely to be affected by a local fluctuation in gas flow and a local surface condition of the housing **20**, and the ignition stability of the ammonia gas is improved. As a result, for example, unlike a case where a discharge is generated in a narrow range only in the vicinity of the tip portion **70***a* of the discharge electrode terminal **70**, it is not necessary to locally adjust the flow speed and the air-fuel ratio of the ammonia gas.
(40) In addition, the discharge electrode terminal **70** is disposed in the region including the inside on the closed wall **60** side of the housing **20**. Therefore, a discharge is generated at a position away from the open end **20***a* of the housing **20**, the open end **20***a* forming the discharge port for the combustion gas. Accordingly, combustion is stabilized inside the housing **20**.
(41) In addition, in the present embodiment, the discharge electrode terminal **70** is attached to the closed wall **60** of the housing **20** via the insulating body **110**. Hence, the discharge electrode

terminal **70** is unlikely to interfere with a tubular flow of the ammonia gas and the air. Therefore, the ammonia gas and the air easily flow inside the housing **20** in a swirling manner, so that the ignition stability of the ammonia gas is more improved.
(42) In addition, in the present embodiment, the discharge electrode terminal **70** is disposed at the radially center portion of the housing **20** inside the housing **20**. Hence, the distance from the tip portion **70***a* of the discharge electrode terminal **70** to the housing **20** is equal over the whole circumference of the housing **20**, so that a discharge is uniformly generated between the tip portion **70***a* of the discharge electrode terminal **70** and the housing **20** along the circumferential direction of the housing **20**. Therefore, the ignition stability of the ammonia gas is more improved.
(43) In addition, in the present embodiment, the tip portion **70***a* of the discharge electrode terminal **70** is located between the rear end **30***b* of the ammonia gas introduction portion **30** and the closed wall **60**, the tip portion **70***a* of the discharge electrode terminal **70** is located between the rear end **400***b* of the air introduction portion **400** and the closed wall **60**. Hence, the discharge electrode terminal **70** is more unlikely to interfere with a tubular flow of the ammonia gas and the air. Therefore, the ammonia gas and the air more easily flow inside the housing **20** in a swirling manner, so that the ignition stability of the ammonia gas is even more improved. In addition, since a discharge is generated at a position sufficiently away from the open end **20***a* of the housing **20**, combustion is more stabilized inside the housing **20**.
(44) FIG. **4** is a configuration view illustrating a modification example of the tubular flame burner **100** illustrated in FIG. **3**. In FIG. **4**, in the tubular flame burner **100** of the present modification example, the discharge electrode terminal **70** is attached to the circular cylindrical portion **20***b* of the housing **20** via the insulating body **110**.
(45) Specifically, the discharge electrode terminal **70** is attached to the circular cylindrical portion **20***b* via the insulating body **110** to penetrate through the circular cylindrical portion **20***b* of the housing **20**. The discharge electrode terminal **70** is disposed to extend in a radial direction of the housing **20**. The tip portion **70***a* of the discharge electrode terminal **70** is located at the radially center portion of the housing **20** between each of the rear end **30***b* of the ammonia gas introduction portion **30** and the rear end **400***b* of the air introduction portion **400** and the closed wall **60**. As described above, in the present modification example, the degree of freedom in the disposition of the discharge electrode terminal **70** is increased.
(46) Incidentally, the present disclosure is not limited to the above embodiment. For example, in the above embodiment, the discharge electrode terminal **70** is disposed at the radially center portion of the housing **20** inside the housing **20**, but the present disclosure is not particularly limited to such a form. For example, depending on the value of a voltage applied to the discharge electrode terminal **70**, the discharge electrode terminal **70** may be disposed to be radially offset from the radially center portion of the housing **20** inside the housing **20**.
(47) In addition, in the above embodiment, the tip portion **70***a* of the discharge electrode terminal **70** is located between the rear end **30***b* of the ammonia gas introduction portion **30** and the closed wall **60**, but the present disclosure is not particularly limited to such a form. The tip portion **70***a* of the discharge electrode terminal **70** may be located between the front end **30***a* and the rear end **30***b* of the ammonia gas introduction portion **30**.
(48) In addition, in the above embodiment, the two ammonia gas introduction portions **30** that introduce the ammonia gas into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**, and the two air introduction portions **400** that introduce the air into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20** are provided, but the number of the ammonia gas introduction portions **30** and of the air introduction portions **400** may be 1 or 3 or more.
(49) In addition, in the above embodiment, the ammonia gas introduction portions **30** and the air introduction portions **400** separately introduce the ammonia gas and the air into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**, but the present

disclosure is not particularly limited to such a form, and at least one introduction portion that introduces a mixed gas of the ammonia gas and the air into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20** may be provided.
(50) In addition, in the above embodiment, the ammonia gas is introduced as a fuel into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**, but the fuel is not particularly limited to the ammonia gas, and may be a fuel gas such as hydrocarbon gas, methane gas, or city gas or may be a liquid fuel or the like that vaporizes at a relatively low temperature, such as liquid ammonia, kerosene, alcohol, or A-type heavy oil.
(51) In addition, in the above embodiment, the air is introduced as an oxidizing gas into the housing **20** in the tangential direction of the inner peripheral surface **20***c* of the housing **20**, but the oxidizing gas is not particularly limited to the air and may be oxygen.
(52) [Second Embodiment] FIG. **5** is a cross-sectional view illustrating a combustor according to a second embodiment of the present disclosure. In FIG. **5**, a combustor **1** of the present embodiment is a tubular flame burner that combusts ammonia gas (NH.sub.3 gas) mixed with air, to generate combustion gas. The ammonia gas is a fuel. The air is an oxidizing gas.
(53) As illustrated in FIG. **6**, the combustor **1** includes a housing **2** having a circular tubular shape, two ammonia gas introduction portions **3** that introduce the ammonia gas into the housing **2**, two air introduction portions **4** that introduce the air into the housing **2**, and an ignition unit **5** that ignites the ammonia gas introduced into the housing **2**.
(54) One end side of the housing **2** is open and the other end side of the housing **2** is closed. One end of the housing **2** forms a gas outlet portion **6** from which combustion gas is discharged. A closed wall **7** is provided at the other end of the housing **2**. The housing **2** and the closed wall **7** are made of a metal material having conductivity (for example, stainless steel). The ammonia gas, the air, and the combustion gas flow inside the housing **2** in an axial direction (direction A) of the housing **2**.
(55) For example, the ammonia gas introduction portions **3** and the air introduction portions **4** are disposed on a closed wall **7** side with respect to a center portion in the axial direction of the housing **2**. As illustrated in FIG. **2**, the ammonia gas introduction portions **3** and the air introduction portions **4** are alternately disposed at equal intervals along a circumferential direction of the housing **2**. The ammonia gas introduction portion **3** and the air introduction portion **4** are introduction portions that introduce the ammonia gas and the air into the housing **2** to generate a tubular flow.
(56) Incidentally, the number of the ammonia gas introduction portions **3** and of the air introduction portions **4** is not particularly limited to 2 and may be 1 or 3 or more. In addition, the ammonia gas introduction portions **3** and the air introduction portions **4** may be provided at the center portion in the axial direction of the housing **2** or may be provided on a gas outlet portion **6** side with respect to the center portion in the axial direction of the housing **2**.
(57) Specifically, the ammonia gas introduction portion **3** introduces the ammonia gas into the housing **2** in a tangential direction of an inner peripheral surface **2***a* of the housing **2**. The air introduction portion **4** introduces the air into the housing **2** in the tangential direction of the inner peripheral surface **2***a* of the housing **2**. The ammonia gas introduction portions **3** and the air introduction portions **4** may be integrally formed with the housing **2** or may be configured separately from the housing **2** and fixed to the housing **2**.
(58) The ignition unit **5** ignites the ammonia gas introduced into the housing **2**, to generate the combustion gas including a tubular flame. The ignition unit **5** includes an ignition plug **8** disposed on the other end side (closed wall **7** side) of the housing **2**, and a voltage supply unit **9** that supplies a voltage to the ignition plug **8**. The ignition plug **8** penetrates through the closed wall **7**. A tip side portion of the ignition plug **8** is disposed inside the housing **2**.
(59) FIG. **7** is a side cut-away view of the ignition plug **8**. FIG. **8** is a schematic front view of the ignition plug **8**. In FIGS. **7** and **8**, the ignition plug **8** is a plug that ignites a mixed gas of the

ammonia gas and the air. The ignition plug **8** includes an insulator **10**, a center electrode **11**, and a ground electrode **12**. Incidentally, in FIG. **5**, the ignition plug **8** is illustrated in a simplified manner. In addition, in FIGS. **7** and **8**, the housing **2** and the closed wall **7** are omitted.

(60) The insulator **10** has a circular cylindrical shape. The insulator **10** is made of ceramic, such as alumina, having good insulation, heat resistance, and thermal conductivity. The insulator **10** is provided with an axial hole **10***a* extending in an axial direction of the insulator **10**.

(61) The center electrode **11** is provided at a tip of a center pole **13** having a round bar shape and being supported by the insulator **10**. The center pole **13** is supported by the insulator **10** in a state where the center pole **13** is inserted into the axial hole **10***a* of the insulator **10**. The center pole **13** is made of, for example, a steel material or the like.

(62) The center electrode **11** protrudes from a tip surface **10***b* of the insulator **10**. Namely, the center electrode **11** is exposed from the tip surface **10***b* of the insulator **10**. The center electrode **11** has a circular shape in a front view. The center electrode **11** is made of, for example, a metal material such as a nickel alloy having good heat resistance and corrosion resistance. Incidentally, a noble metal tip may be provided on a tip surface **11***a* of the center electrode **11**.

(63) The ground electrode **12** is integrated with a metal shell **14** disposed around the insulator **10**. The metal shell **14** is fixed to an outer peripheral surface of the insulator **10**. The metal shell **14** has a circular cylindrical shape. Incidentally, the circular cylindrical shape referred to in the present embodiment is not limited to a perfect circular cylindrical shape and also includes a substantially circular cylindrical shape.

(64) The ground electrode **12** is joined to a tip of the metal shell **14** by welding or the like. The ground electrode **12** is an electrode having a circular cylindrical shape and being disposed around the insulator **10**. The ground electrode **12** is made of a metal material such as a nickel alloy having good heat resistance and corrosion resistance. The ground electrode **12** is grounded.

(65) The ground electrode **12** is fixed to the closed wall **7**. A male screw portion **12***a* to be screwed to a female screw portion **7***a* (refer to FIG. **21**) provided in the closed wall **7** is formed in an outer peripheral surface of the ground electrode **12**. The ignition plug **8** including the ground electrode **12** is fixed to the closed wall **7** by screwing the ground electrode **12** into the closed wall **7**. Incidentally, the ignition plug **8** is fixed to the closed wall **7** such that the center electrode **11** is located at a radially center portion of the housing **2** inside the housing **2**.

(66) The metal shell **14** is fixed to the housing **2** via the ground electrode **12** and via the closed wall **7**. At this time, an axial direction of the ground electrode **12** and the metal shell **14** coincides with the axial direction (direction A in FIG. **5**) of the housing **2**. Incidentally, the coincidence of the axial directions referred to in the present embodiment is not limited to a perfect coincidence and also includes a coincidence by appearance.

(67) The ground electrode **12** is disposed outside the center electrode **11** in a radial direction so as not to overlap the center electrode **11** in the axial direction of the ground electrode **12**. Namely, the ground electrode **12** is disposed at a constant interval from the center electrode **11** in the radial direction. The tip surface **11***a* of the center electrode **11** protrudes from a tip surface **12***b* of the ground electrode **12**. Namely, the tip surface **11***a* of the center electrode **11** is located on the one end side (gas outlet portion **6** side) of the housing **2** with respect to the tip surface **12***b* of the ground electrode **12**. A space S that the ammonia gas and the air reach is provided between the center electrode **11** and the ground electrode **12**.

(68) A base end portion of the ignition plug **8** is provided with a terminal metal fixture **15**. The terminal metal fixture **15** is electrically connected to the center pole **13**. The terminal metal fixture **15** is exposed from a base end surface **10***c* of the insulator **10** to be located outside the housing **2**.

(69) As illustrated in FIG. **5**, the voltage supply unit **9** is connected to the terminal metal fixture **15** of the ignition plug **8** via a high-voltage cable **16**. The voltage supply unit **9** applies a high voltage to the center electrode **11** via the high-voltage cable **16**, via the terminal metal fixture **15**, and via the center pole **13**.

(70) The center electrode **11** functions as a discharge electrode. The ignition unit **5** ignites the ammonia gas by causing the voltage supply unit **9** to apply a high voltage to the center electrode **11** and thus generating a discharge in the space S between the center electrode **11** and the ground electrode **12** in the ignition plug **8**. At this time, the discharge is generated in a region where the distance between a tip portion of the center electrode **11** and the ground electrode **12** is at its shortest in the space S between the tip portion of the center electrode **11** and the ground electrode **12**.
(71) In the combustor **1** as described above, when the ammonia gas introduction portions **3** introduce the ammonia gas into the housing **2** in the tangential direction of the inner peripheral surface **2***a* of the housing **2**, and the air introduction portions **4** introduce the air into the housing **2** in the tangential direction of the inner peripheral surface **2***a* of the housing **2**, the ammonia gas and the air form a tubular flow and are mixed and flow inside the housing **2** in a swirling manner. At this time, a mixed gas of the ammonia gas and the air flows inside the housing **2** toward the gas outlet portion **6** and flows inside the housing **2** toward the ignition plug **8**.
(72) In that state, when a power source of the voltage supply unit **9** is turned on, the voltage supply unit **9** applies a high voltage to the center electrode **11** of the ignition plug **8** via the terminal metal fixture **15** and via the center pole **13**. Then, as illustrated in FIG. **9**, a discharge P is generated in the space S between the tip portion of the center electrode **11** and the tip surface **12***b* of the ground electrode **12** in the ignition plug **8**. At this time, the center electrode **11** protrudes from the tip surface **10***b* of the insulator **10**. For this reason, the discharge P generated in the space S between the center electrode **11** and the ground electrode **12** is easily separated from the insulator **10**.
(73) The ammonia gas is ignited and combusted by the discharge P, to form a tubular flame, so that high-temperature combustion gas is generated. The high-temperature combustion gas flows inside the housing **2** toward the gas outlet portion **6** and is discharged from the gas outlet portion **6**.
(74) FIG. **10** is a cross-sectional view illustrating a combustor as a comparative example. In FIG. **10**, a combustor **1**A of the present comparative example includes an ignition plug **101** for an internal combustion engine. The ignition plug **101** includes the insulator **10** and the center electrode **11** that are the same as those of the ignition plug **8**, and a ground electrode **102**.
(75) The ground electrode **102** is integrated with a metal shell **103** having a circular cylindrical shape and being disposed around the insulator **10**. A male screw portion **103***a* to be screwed into a screw hole (not illustrated) of an internal combustion engine is provided in an outer peripheral surface of a tip side portion of the metal shell **103**. The ground electrode **102** is joined to a tip of the metal shell **103** by welding or the like. The ground electrode **102** is bent in an L shape toward a center electrode **11** side (radially center side of the ground electrode **102**). A space between the center electrode **11** and a tip portion **102***a* of the ground electrode **102** is a discharge gap **104** at which a discharge is generated. The ammonia gas and the air flow into the discharge gap **104** in a radial direction of the housing **2**.
(76) By the way, the ammonia gas and the air are introduced into the housing **2** in the tangential direction of the inner peripheral surface **2***a* of the housing **2** and flow as a tubular flow inside the housing **2** in a swirling manner. The introduction positions of the ammonia gas and the air are positions in the vicinity of the ignition plug **101**. For this reason, in the vicinity of the ignition plug **101**, the flow speed of the ammonia gas and the air is higher on a radially outer side of the housing **2** than a radially inner side (center side) of the housing **2**. At this time, a phenomenon that a flow (backflow) of the ammonia gas and the air toward the ignition plug **101** occurs in a radially inner region of the housing **2** inside the housing **2** before the ignition of the ammonia gas has been confirmed by a simulation.
(77) Since the backflow of the ammonia gas and the air is a flow in the axial direction of the housing **2**, the ground electrode **102** interferes with the flow of the ammonia gas and the air to the discharge gap **104**. For this reason, the ammonia gas and the air to flow are unlikely to flow into the discharge gap **104**. Therefore, a discharge generated in the discharge gap **104** is unlikely to act

on a mixed gas of the ammonia gas and the air, so that it is difficult to stabilize an operation of igniting the ammonia gas and causing a tubular flame to grow.

(78) In response to such a problem, in the present embodiment, the ground electrode **12** is integrated with the metal shell **14** disposed around the insulator **10**, and is disposed outside the center electrode **11** in a radial direction of the metal shell **14** so as not to overlap the center electrode **11** in the axial direction of the metal shell **14**. For this reason, the ground electrode **12** is prevented from interfering with a flow of the ammonia gas and the air to the space S between the center electrode **11** and the ground electrode **12** even when a flow of the ammonia gas and the air toward the ignition plug **8** is generated in the radially inner (center side) region of the housing **2** inside the housing **2** by a tubular flow of the ammonia gas and the air. Therefore, the ammonia gas and the air easily flow into the space S between the center electrode **11** and the ground electrode **12**, so that a discharge generated in the space S between the center electrode **11** and the ground electrode **12** easily acts on a mixture gas of the ammonia gas and the air. In addition, after the ammonia gas is ignited by the discharge, the flame burns and spreads to the surrounding mixed gas, but since the mixed gas flows as a tubular flow to the one end side of the housing **2** in the axial direction of the metal shell **14**, the tubular flame expands in the axial direction. At this time, the ground electrode **12** is disposed not to overlap the center electrode **11** in the axial direction of the metal shell **14**. For this reason, when the ammonia gas is ignited, heat is unlikely to be taken away from the tubular flame by the ground electrode **12** and the tubular flame easily expands. As described above, the ignition and combustion operation of the ammonia gas is stabilized.

(79) In addition, in the present embodiment, the ground electrode **12** has a circular cylindrical shape and is integrated with the metal shell **14** so as to be disposed around the insulator **10**. Therefore, a structure of the existing ground electrode can be used.

(80) In addition, in the present embodiment, the tip surface **11***a* of the center electrode **11** is located on the one end side (gas outlet portion **6** side) of the housing **2** with respect to the ground electrode **12**. For this reason, the ammonia gas and the air easily flow from a center electrode **11** side to a ground electrode **12** side. Therefore, the ammonia gas and the air more easily flow into the space S between the center electrode **11** and the ground electrode **12**.

(81) [Third Embodiment] FIG. **11** is a side cut-away view illustrating an ignition plug in a combustor according to a third embodiment of the present disclosure. FIG. **12** is a schematic front view of the ignition plug illustrated in FIG. **11**. In FIGS. **11** and **12**, in the combustor **1** of the present embodiment, the ignition plug **8** includes a center electrode **21** instead of the center electrode **11** in the second embodiment.

(82) The center electrode **21** has a circular shape in a front view. A tip surface **21***a* of the center electrode **21** is located on the one end side (gas outlet portion **6** side) of the housing **2** with respect to the tip surface **12***b* of the ground electrode **12**. A diameter D**1** of the center electrode **21** is larger than a diameter D**2** of the tip surface **10***b* of the insulator **10** and smaller than a diameter D**3** (outer diameter) of the tip surface **12***b* of the ground electrode **12**.

(83) Therefore, the center electrode **21** includes a protrusion portion **22** protruding outward in a radial direction of the insulator **10** with respect to a peripheral edge **10***d* of the tip surface **10***b* of the insulator **10**. A side end **22***a* of the protrusion portion **22** is located in a region outside the peripheral edge **10***d* of the tip surface **10***b* of the insulator **10** in the radial direction of the insulator **10** and inside an outer periphery of a tip portion of the ground electrode **12** in a radial direction of the ground electrode **12**. The protrusion portion **22** is provided at a peripheral edge portion of the center electrode **21** and has an annular shape.

(84) In the combustor **1** including the ignition plug **8**, when the voltage supply unit **9** applies a high voltage to the center electrode **21** of the ignition plug **8** via the terminal metal fixture **15** and via the center pole **13**, the discharge P is generated in the space S between the protrusion portion **22** of the center electrode **21** and the tip surface **12***b* of the ground electrode **12** in the ignition plug **8**, and the ammonia gas is ignited and combusted.

(85) As described above, in the present embodiment, when a voltage is supplied to the center electrode **21**, a discharge is generated in the space S between the protrusion portion **22** of the center electrode **21** and the ground electrode **12**. Therefore, the discharge is generated in the space separated from the insulator **10**, so that heat caused by the ignition of the ammonia gas is more unlikely to be taken away by the insulator **10**. Accordingly, the ignition and combustion operation of the ammonia gas is more stabilized.
(86) In addition, in the present embodiment, the protrusion portion **22** is provided at the peripheral edge portion of the center electrode **21** having a circular shape and has an annular shape. For this reason, a discharge can be generated in the space S between the protrusion portion **22** and the ground electrode **12** over the whole circumference of the center electrode **21**. In addition, since the shape of the center electrode **21** is a circular shape, the center electrode **21** can be easily manufactured.
(87) FIG. **13** is a schematic front view of the ignition plug **8** including a modification example of the center electrode **21** illustrated in FIG. **12**. In the ignition plug **8** illustrated in (a) in FIG. **13**, the shape of the center electrode **21** is an oblong shape in a front view. Each of both end portions in a longitudinal direction of the center electrode **21** is provided with the protrusion portion **22** protruding in the radial direction of the insulator **10** with respect to the peripheral edge **10***d* of the tip surface **10***b* of the insulator **10**.
(88) In the ignition plug **8** illustrated in (b) in FIG. **13**, the shape of the center electrode **21** is a cross shape in a front view. Each of end portions of the cross of the center electrode **21** is provided with the protrusion portion **22** protruding in the radial direction of the insulator **10** with respect to the peripheral edge **10***d* of the tip surface **10***b* of the insulator **10**.
(89) In the ignition plug **8** illustrated in (c) in FIG. **13**, the shape of the center electrode **21** is a polygonal shape (here, a hexagonal shape) in a front view. The peripheral edge portion of the center electrode **21** is provided with the protrusion portion **22** having an angled annular shape (here, a hexagonal annular shape) and protruding in the radial direction of the insulator **10** with respect to the peripheral edge **10***d* of the tip surface **10***b* of the insulator **10**.
(90) In any one of (a) to (c) in FIGS. **13** to **13**, when a high voltage is applied to the center electrode **21** of the ignition plug **8**, a discharge is generated in the space S between the protrusion portion **22** of the center electrode **21** and the ground electrode **12**, and the ammonia gas is ignited and combusted. Incidentally, the shape of the center electrode **21** is not limited to the shapes illustrated in FIG. **13**.
(91) [Fourth Embodiment] FIG. **14** is a side cut-away view illustrating an ignition plug in a combustor according to a fourth embodiment of the present disclosure. FIG. **15** is a schematic front view of the ignition plug illustrated in FIG. **14**. In FIGS. **14** and **15**, in the combustor **1** of the present embodiment, the ignition plug **8** includes a ground electrode **32** instead of the ground electrode **12** in the second embodiment.
(92) The ground electrode **32** includes an electrode main body **33** having a circular cylindrical shape and corresponding to the ground electrode **12** in the second embodiment, and two projections **34** each having a plate shape and being integrated with the electrode main body **33**. A male screw portion **33***a* is formed in an outer peripheral surface of the electrode main body **33**.
(93) The projections **34** protrude from a tip surface **33***b* of the electrode main body **33**. Namely, a tip portion of the ground electrode **32** is provided with the two projections **34** protruding to the one end side (gas outlet portion **6** side) of the housing **2**. For example, the two projections **34** are disposed to face each other with the center electrode **11** interposed between. The tip surface **11***a* of the center electrode **11** is located on the one end side of the housing **2** with respect to tip surfaces **34***a* of the projections **34**.
(94) In the combustor **1** including the ignition plug **8**, when the voltage supply unit **9** applies a high voltage to the center electrode **11** of the ignition plug **8** via the terminal metal fixture **15** and via the center pole **13**, the discharge P is generated in the space S between the tip portion of the center

electrode **11** and tip portions of the projections **34** of the ground electrode **32** in the ignition plug **8**, and the ammonia gas is ignited and combusted.

(95) In the present embodiment as described above, when a voltage is supplied to the center electrode **11**, a discharge is generated in the space S between the center electrode **11** and the projections **34**. Therefore, the discharge is generated in the space separated from the insulator **10**, so that heat caused by the ignition of the ammonia gas is more unlikely to be taken away by the insulator **10**. Accordingly, the ignition and combustion operation of the ammonia gas is more stabilized.

(96) FIG. **16** is a side cut-away view of the ignition plug **8** including a modification example of the ground electrode **32** illustrated in FIG. **15**. In FIG. **16**, bent portions **35** mutually bent at an obtuse angle to the center electrode **11** side are provided on tip sides of the projections **34** of the ground electrode **32**. In this case, since the distance between the center electrode **11** and the projection **34** is short, the discharge P is easily generated.

(97) Incidentally, in the present embodiment and the modification examples, the tip portion of the ground electrode **32** is provided with the two projections **34**, but the number of the projections **34** is not particularly limited to 2 and may be 1 or 3 or more.

(98) [Fifth Embodiment] FIG. **17** is a side cut-away view illustrating an ignition plug in a combustor according to a fifth embodiment of the present disclosure. In FIG. **17**, in the combustor **1** of the present embodiment, similarly to the fourth embodiment, the ignition plug **8** includes the ground electrode **32**. The tip surface **34***a* of the projection **34** of the ground electrode **32** is located on the one end side (gas outlet portion **6** side) of the housing **2** with respect to the tip surface **11***a* of the center electrode **11**.

(99) In the combustor **1** including the ignition plug **8**, when the voltage supply unit **9** applies a high voltage to the center electrode **11** of the ignition plug **8** via the terminal metal fixture **15** and via the center pole **13**, the discharge P is generated in the space S between the tip portion of the center electrode **11** and side surfaces **34***b* on an inner side (center electrode **11** side) of the projections **34** in the ignition plug **8**, and the ammonia gas is ignited and combusted.

(100) In the present embodiment as described above, when a voltage is supplied to the center electrode **11**, a discharge is generated in the space S between the center electrode **11** and the side surfaces **34***b* of the projections **34**. Therefore, the discharge is generated in the space sufficiently separated from the insulator **10**, so that heat caused by the ignition of the ammonia gas is more unlikely to be taken away by the insulator **10**. Accordingly, the ignition and combustion operation of the ammonia gas is even more stabilized.

(101) [Sixth Embodiment] FIG. **18** is a side cut-away view illustrating an ignition plug in a combustor according to a sixth embodiment of the present disclosure. In FIG. **18**, in the combustor **1** of the present embodiment, the ignition plug **8** includes a ground electrode **40** instead of the ground electrode **12** in the second embodiment. The ground electrode **40** is disposed outside the center electrode **11** in the radial direction of the metal shell **14** so as not to overlap the center electrode **11** in the axial direction of the metal shell **14**.

(102) The ground electrode **40** includes a circular cylindrical portion **41** corresponding to the ground electrode **12** in the second embodiment, the closed wall **7**, and an erected portion **43** having a bar shape. The closed wall **7** forms an annular portion that is a part of the ground electrode **40**. A male screw portion **41***a* to be screwed to the female screw portion **7***a* of the closed wall **7** is formed in an outer peripheral surface of the circular cylindrical portion **41**.

(103) The erected portion **43** is integrated with the closed wall **7**. The erected portion **43** is provided on a tip surface **7***b* of the closed wall **7** to extend to the one end side (gas outlet portion **6** side) of the housing **2**. The erected portion **43** is disposed outside the circular cylindrical portion **41** in a radial direction of the closed wall **7**. Incidentally, the number of the erected portions **43** may be 1 or plural.

(104) A tip surface **40***a* of the ground electrode **40** is located on the one end side of the housing **2**

with respect to the tip surface **11***a* of the center electrode **11**. The tip surface **40***a* of the ground electrode **40** corresponds to a tip surface **43***a* of the erected portion **43**. A distance between the center electrode **11** and a peripheral surface **43***b* of the erected portion **43** is shorter than a distance between the center electrode **11** and a tip surface **41***b* of the circular cylindrical portion **41**. The distance between the center electrode **11** and the peripheral surface **43***b* of the erected portion **43** is a distance along the radial direction of the metal shell **14**.

(105) In the combustor **1** including the ignition plug **8** described above, when the voltage supply unit **9** applies a high voltage to the center electrode **11** of the ignition plug **8** via the terminal metal fixture **15** and via the center pole **13**, the discharge P is generated in the space S between the tip portion of the center electrode **11** and the peripheral surface **43***b* of the erected portion **43** in the ignition plug **8**, and the ammonia gas is ignited and combusted.

(106) In the present embodiment as described above, when a voltage is supplied to the center electrode **11**, a discharge is generated in the space S between the center electrode **11** and the erected portion **43**. Therefore, the discharge is generated in the space separated from the insulator **10**, so that heat caused by the ignition of the ammonia gas is more unlikely to be taken away by the insulator **10**. Accordingly, the ignition and combustion operation of the ammonia gas is more stabilized. In addition, the discharge energy can be reduced by reducing the heat capacity of the erected portion **43** compared to the circular cylindrical portion **41**.

(107) FIG. **19** is a front view of the ignition plug **8** including a modification example of the ground electrode **40** illustrated in FIG. **18**. In FIG. **19**, in the present modification example, a tip side portion of the erected portion **43** of the ground electrode **40** is provided with a bent portion **44** bent at an obtuse angle to the center electrode **11** side. In this case, since the distance between the center electrode **11** and the erected portion **43** is short, the discharge is easily generated.

(108) [Seventh Embodiment] FIG. **20** is a side cut-away view illustrating an ignition plug in a combustor according to a seventh embodiment of the present disclosure. In FIG. **20**, in the combustor **1** of the present embodiment, the ignition plug **8** includes a ground electrode **45** instead of the ground electrode **40** in the sixth embodiment.

(109) The ground electrode **45** includes the circular cylindrical portion **41**, the closed wall **7**, a plurality (here, two) of erected portions **46** each having a bar shape, and a connection bar **47**. The erected portions **46** are provided on the tip surface **7***b* of the closed wall **7** to extend to the one end side (gas outlet portion **6** side) of the housing **2**. The connection bar **47** is a connection portion connecting tips of each of the erected portions **46**. A tip surface **45***a* of the ground electrode **45** corresponds to a tip surface **47***a* of the connection bar **47**.

(110) A distance R**1** between the center electrode **11** and the erected portion **46** is shorter than a distance R**2** between the center electrode **11** and the connection bar **47**. The distance R**1** between the center electrode **11** and the erected portion **46** is a distance along the radial direction of the metal shell **14**. The distance R**2** between the center electrode **11** and the connection bar **47** is a distance along the axial direction of the metal shell **14**.

(111) In the combustor **1** including the ignition plug **8**, when the voltage supply unit **9** applies a high voltage to the center electrode **11** of the ignition plug **8** via the terminal metal fixture **15** and via the center pole **13**, the discharge P is generated in the space S between the tip portion of the center electrode **11** and peripheral surfaces **46***b* of the erected portions **46** in the ignition plug **8**, and the ammonia gas is ignited and combusted.

(112) In the present embodiment as described above, the tips of the plurality of erected portions **46** are connected by the connection bar **47**, thereby leading to an increase in the strength of the erected portions **46**. In addition, when a voltage is supplied to the center electrode **11**, the discharge P is generated in the space S between the center electrode **11** and the erected portions **46**, and the discharge P is not generated in a space between the center electrode **11** and the connection bar **47**. In addition, the discharge energy can be reduced by reducing the heat capacity of the erected portion **46** and the connection bar **47** compared to the circular cylindrical portion **41**.

(113) [Eighth Embodiment] FIG. **21** is a side cut-away view illustrating an ignition plug in a combustor according to an eighth embodiment of the present disclosure. In FIG. **21**, in the combustor **1** of the present embodiment, the ignition plug **8** includes a ground electrode **50** instead of the ground electrode **12** in the second embodiment. The ground electrode **50** is disposed outside the center electrode **11** in the radial direction of the metal shell **14** so as not to overlap the center electrode **11** in the axial direction of the metal shell **14**.
(114) The ground electrode **50** includes the circular cylindrical portion **41** that is the same as that of the sixth embodiment, and a protrusion **51** provided on the inner peripheral surface **2***a* of the housing **2**. The protrusion **51** is provided on the inner peripheral surface **2***a* of the housing **2** to protrude inward in the radial direction of the housing **2** toward the center electrode **11**. The protrusion **51** extends from the inner peripheral surface **2***a* of the housing **2** to the vicinity of the center electrode **11**. The number of the protrusions **51** may be 1 or plural. The shape of the protrusion **51** is, for example, a rectangular shape, a fan shape, an annular shape, or the like when viewed in the axial direction of the metal shell **14**.
(115) A side surface **51***a* on the gas outlet portion **6** side of the protrusion **51** is located on the gas outlet portion **6** side (one end side of the housing **2**) with respect to the tip surface **11***a* of the center electrode **11**. A distance between the center electrode **11** and a tip surface **51***b* of the protrusion **51** is shorter than a distance between the center electrode **11** and the tip surface **41***b* of the circular cylindrical portion **41**.
(116) In the combustor **1** including the ignition plug **8**, when the voltage supply unit **9** applies a high voltage to the center electrode **11** of the ignition plug **8** via the terminal metal fixture **15** and via the center pole **13**, the discharge P is generated in the space S between the tip portion of the center electrode **11** and the tip surface **51***b* of the protrusion **51** in the ignition plug **8**, and the ammonia gas is ignited and combusted.
(117) In the present embodiment as described above, when a voltage is supplied to the center electrode **11**, a discharge is generated in the space S between the center electrode **11** and the protrusion **51**. Therefore, the discharge is generated in the space separated from the insulator **10**, so that heat caused by the ignition of the ammonia gas is more unlikely to be taken away by the insulator **10**. Accordingly, the ignition and combustion operation of the ammonia gas is more stabilized. In addition, the structure of the ground electrode **50** can be simplified. Further, the discharge energy can be reduced by reducing the heat capacity of the protrusion **51** compared to the circular cylindrical portion **41**.
(118) Several embodiments of the present disclosure have been described above, but the present disclosure is not limited to the above embodiments. For example, in the second and third embodiments, the tip surfaces **11***a* and **21***a* of the center electrodes **11** and **21** are located on the one end side (gas outlet portion **6** side) of the housing **2** with respect to the tip surface **12***b* of the ground electrode **12**, but the present disclosure is not particularly limited to such a form. The tip surfaces **11***a* and **21***a* of the center electrodes **11** and **21** may be located to be flush with the tip surface **12***b* of the ground electrode **12** or may be located on the other end side (closed wall **7** side) of the housing **2** with respect to the tip surface **12***b* of the ground electrode **12**.
(119) In addition, in the sixth embodiment, the tip surface **40***a* of the ground electrode **40** is located on the one end side of the housing **2** with respect to the tip surface **11***a* of the center electrode **11**, but the present disclosure is not particularly limited to such a form. The tip surface **40***a* of the ground electrode **40** may be located to be flush with the tip surface **11***a* of the center electrode **11** or may be located on the other end side of the housing **2** with respect to the tip surface **11***a* of the center electrode **11**.
(120) In addition, in the eighth embodiment, the side surface **51***a* on the gas outlet portion **6** side of the protrusion **51** in the ground electrode **50** is located on the gas outlet portion **6** side (one end side of the housing **2**) with respect to the tip surface **11***a* of the center electrode **11**, but the present disclosure is not particularly limited to such a form. The side surface **51***a* on the gas outlet portion

6 side of the protrusion **51** may be located to be flush with the tip surface **11***a* of the center electrode **11** or may be located on the other end side of the housing **2** with respect to the tip surface **11***a* of the center electrode **11**.

(121) In addition, in the sixth to eighth embodiments, each of the ground electrodes **40**, **45**, and **50** includes the circular cylindrical portion **41** corresponding to the ground electrode **12** in the first embodiment, but the circular cylindrical portion **41** may not be particularly provided.

(122) In addition, in the above embodiments, the metal shell **14** has a circular cylindrical shape, but the shape of the metal shell **14** is not particularly limited to a circular cylindrical shape and may be a cylindrical shape. In this case, the cylindrical shape is not limited only to a perfect cylindrical shape and also includes a substantially cylindrical shape.

(123) In addition, in the above embodiments, the ammonia gas introduction portions **3** introduce the ammonia gas into the housing **2** in the tangential direction of the inner peripheral surface **2***a* of the housing **2**, and the air introduction portions **4** introduce the air into the housing **2** in the tangential direction of the inner peripheral surface **2***a* of the housing **2**, but the present disclosure is not particularly limited to such a form. The ammonia gas introduction portions **3** may introduce the ammonia gas into the housing **2** so as to be offset from the tangential direction of the inner peripheral surface **2***a* of the housing **2** as long as the ammonia gas introduction portions **3** introduce the ammonia gas into the housing **2** to generate a tubular flow. The air introduction portions **4** may introduce the air into the housing **2** so as to be offset from the tangential direction of the inner peripheral surface **2***a* of the housing **2** as long as the air introduction portions **4** introduce the air into the housing **2** to generate a tubular flow.

(124) In addition, in the above embodiments, the ammonia gas introduction portions **3** and the air introduction portions **4** separately introduce the ammonia gas and the air into the housing **2** to generate a tubular flow, but the present disclosure is not particularly limited to such a form, and at least one introduction portion that introduces a mixed gas of the ammonia gas and the air into the housing **2** to generate a tubular flow may be provided.

(125) In addition, in the above embodiments, the ammonia gas is introduced as a fuel into the housing **2** to generate a tubular flow, but the fuel is not particularly limited to the ammonia gas, and may be a fuel gas such as hydrocarbon gas, methane gas, or city gas or may be a liquid fuel or the like that vaporizes at a relatively low temperature, such as liquid ammonia, kerosene, alcohol, or A-type heavy oil.

(126) In addition, in the above embodiments, the air is introduced as an oxidizing gas into the housing **2** to generate a tubular flow, but the oxidizing gas is not particularly limited to the air and may be oxygen.

REFERENCE SIGNS LIST

(127) **100**, **1**: tubular flame burner (combustor), **20**, **2**: housing, **20***a*: open end, **20***c*, **2***a*: inner peripheral surface, **30**, **3**: ammonia gas introduction portion (introduction portion), **30***b*: rear end (end on closed wall side), **400**, **4**: air introduction portion (introduction portion), **400***b*: rear end (end on closed wall side), **500**, **5**: ignition unit, **60**, **7**: closed wall (annular portion), **70**: discharge electrode terminal, **140**: voltage supply unit, **110**: insulating body, **8**: ignition plug, **10**: insulator, **10***b*: tip surface, **10***d*: peripheral edge, **11**: center electrode, **11***a*: tip surface (tip), **12**: ground electrode, **13**: center pole, **14**: metal shell, **21**: center electrode, **21***a*: tip surface (tip), **22**: protrusion portion, **22***a*: side end, **32**: ground electrode, **34**: projection, **34***a*: tip surface (tip), **40**: ground electrode, **43**: erected portion, **45**: ground electrode, **46**: erected portion, **47**: connection bar (connection portion), **50**: ground electrode, **51**: protrusion, S: space, D**1** to D**3**: diameter, R**1**, R**2**: distance.

Claims

1. A combustor comprising: a housing having a cylindrical shape of which one end side is open and of which the other end side is closed; at least one introduction portion that introduces a fuel and an oxidizing gas into the housing to generate a tubular flow; and an ignition unit that ignites the fuel introduced into the housing, wherein the ignition unit includes a discharge electrode and a ground electrode, and a space that the fuel and the oxidizing gas reach is provided between the discharge electrode and the ground electrode, wherein, a first end of the housing forms an open end of the housing in an axial direction of the housing, and a wall is provided at a second end of the housing that forms a closed end of the housing in the axial direction of the housing, the housing is grounded to function as the ground electrode, the ignition unit includes a discharge electrode terminal, which is disposed at the wall of the closed end of the housing so as to extend inside of the housing, to function as the discharge electrode, and a voltage supply unit that supplies a voltage to the discharge electrode terminal, to generate a discharge between a tip portion of the discharge electrode terminal and the housing to ignite the fuel, the introduction portion is provided between the wall of the housing and the center portion of the housing in the axial direction of the housing on the outer peripheral surface of the housing having the cylindrical shape, and the tip portion of the discharge electrode terminal of the ignition unit is positioned inside the housing between a rear end of the introduction portion and the wall of the closed end of the housing in the axial direction.
2. The combustor according to claim 1, wherein the discharge electrode terminal is attached to the closed wall via an insulating body.
3. The combustor according to claim 2, wherein the discharge electrode terminal is disposed at a radially center portion of the housing inside the housing.
4. A combustor comprising: a housing having a cylindrical shape of which one end side is open and of which the other end side is closed; at least one introduction portion that introduces a fuel and an oxidizing gas into the housing to generate a tubular flow; and an ignition unit that ignites the fuel introduced into the housing, wherein the ignition unit includes a discharge electrode and a ground electrode, and a space that the fuel and the oxidizing gas reach is provided between the discharge electrode and the ground electrode, wherein in the combustor that combusts the fuel mixed with the oxidizing gas, to generate combustion gas, the fuel, the oxidizing gas, and the combustion gas flow in an axial direction of the housing, the ignition unit includes an ignition plug disposed on the other end side of the housing, and ignites the fuel introduced into the housing, to generate the combustion gas including a tubular flame, the ignition plug includes an insulator, a center electrode that is provided at a tip of a center pole supported by the insulator, to function as the discharge electrode, and the ground electrode integrated with a metal shell having a cylindrical shape and being disposed around the insulator, the ignition unit ignites the fuel by supplying a voltage to the center electrode and thus generating a discharge in the space between the center electrode and the ground electrode, the center electrode protrudes from a tip surface of the insulator, and the ground electrode is disposed outside the center electrode in a radial direction of the metal shell so as not to overlap the center electrode in an axial direction of the metal shell, wherein the ground electrode has a cylindrical shape and is integrated with the metal shell so as to be disposed around the insulator.
5. The combustor according to claim 4, wherein a tip of the center electrode is located on the one end side of the housing with respect to the ground electrode.
6. The combustor according to claim 4, wherein the center electrode includes a protrusion portion protruding in a radial direction of the ground electrode with respect to a peripheral edge of the tip surface of the insulator, and a side end of the protrusion portion is located in a region outside the tip surface of the insulator in the radial direction of the ground electrode and inside the ground electrode in the radial direction of the ground electrode.
7. The combustor according to claim 6, wherein the center electrode has a circular shape, a diameter of the center electrode is larger than a diameter of the tip surface of the insulator and

smaller than a diameter of the ground electrode, and the protrusion portion is provided at a peripheral edge portion of the center electrode and has an annular shape.

8. The combustor according to claim 4, wherein a tip portion of the ground electrode is provided with a projection protruding to the one end side of the housing.

9. The combustor according to claim 8, wherein a tip of the projection is located on the one end side of the housing with respect to the center electrode.

10. A combustor comprising: a housing having a cylindrical shape of which one end side is open and of which the other end side is closed; at least one introduction portion that introduces a fuel and an oxidizing gas into the housing to generate a tubular flow; and an ignition unit that ignites the fuel introduced into the housing, wherein the ignition unit includes a discharge electrode and a ground electrode, and a space that the fuel and the oxidizing gas reach is provided between the discharge electrode and the ground electrode, wherein in the combustor that combusts the fuel mixed with the oxidizing gas, to generate combustion gas, the fuel, the oxidizing gas, and the combustion gas flow in an axial direction of the housing, the ignition unit includes an ignition plug disposed on the other end side of the housing, and ignites the fuel introduced into the housing, to generate the combustion gas including a tubular flame, the ignition plug includes an insulator, a center electrode that is provided at a tip of a center pole supported by the insulator, to function as the discharge electrode, and the ground electrode integrated with a metal shell having a cylindrical shape and being disposed around the insulator, the ignition unit ignites the fuel by supplying a voltage to the center electrode and thus generating a discharge in the space between the center electrode and the ground electrode, the center electrode protrudes from a tip surface of the insulator, and the ground electrode is disposed outside the center electrode in a radial direction of the metal shell so as not to overlap the center electrode in an axial direction of the metal shell, wherein the ground electrode includes an annular portion attached to the metal shell, and an erected portion provided on a tip surface of the annular portion to extend to the one end side of the housing.

11. The combustor according to claim 10, wherein a plurality of the erected portions are provided on the tip surface of the annular portion, the ground electrode further includes a connection portion connecting tips of the plurality of erected portions, a tip of the connection portion is located on the one end side of the housing with respect to the center electrode, and a distance between the center electrode and the erected portion is shorter than a distance between the center electrode and the connection portion.

12. The combustor according to claim 4, wherein the metal shell is fixed to the housing, and the ground electrode includes a protrusion provided on an inner peripheral surface of the housing to protrude inward in a radial direction of the housing toward the center electrode.